The Insulin Crew Relay

Model how insulin and working muscles move sugar out of the blood — and why two grabbers at once empty it fast.

Ages 6–9 · 20–25 min · Book link: Chapter 6 and Lab Notes, pages 40–41

YOU NEED

20 paper "sugar cubes" (printable or crumpled paper), three boxes labeled BLOOD, CELLS and LEG MUSCLES, role cards, a timer. Signs, role cards and cubes are on the next pages.

HOW TO PLAY

1. Hand out roles: four Insulin Crew (IRIS, IGGY, INA, INKY), one Brain, two Leg Muscles, and everyone else as the Pancreas Alarm.
2. The rule for the Brain: there must be at least 2 cubes in BLOOD. If there are fewer, the Brain says "foggy…" and everyone wobbles like wet spaghetti for five seconds.
3. The Pancreas Alarm shouts ALARM! whenever BLOOD holds more than 5 cubes, and ALL CLEAR when it's back to 5 or fewer. The Crew may only carry cubes (one at a time, walking) from BLOOD to CELLS while the alarm is on.
4. Round 1 — toast and egg first. The grown-up drops 2 cubes into BLOOD every 10 seconds for one minute. Watch the alarm come and go.
5. Round 2 — three glasses, empty stomach. Dump 15 cubes in at once. Emergency mode: the Crew keeps carrying until the alarm team counts to 10 after ALL CLEAR. What happens to the Brain's cubes?
6. Round 3 — kickoff. Same dump, but now the Leg Muscles grab cubes from BLOOD too — they don't wait for the alarm. How fast does BLOOD empty now?
7. Talk: which round emptied BLOOD fastest? Who were the two grabbers? What could Jiwoo change first?

For the grown-up

It's a model, so it simplifies: insulin doesn't really carry sugar — it signals cells to take it in. (Lumi admits it on page 40: "The Crew is not really a crew.") If a player lives with diabetes, follow their care team's guidance and let them choose their role; never ask them to explain their condition.

Nobody in this game is the villain. Not the sugar, not the Crew, not your legs. — L.

Who's who in your body

IRIS

INSULIN CREW



Carry one cube at a time from BLOOD to CELL only while the alarm is on. Walk, don't run.

IGGY

INSULIN CREW



Carry one cube at a time from BLOOD to CELL only while the alarm is on. Walk, don't run.

INA

INSULIN CREW



Carry one cube at a time from BLOOD to C only while the alarm is on. Walk, don't ru

INKY

INSULIN CREW



Carry one cube at a time from BLOOD to CEL only while the alarm is on. Walk, don't run.

BRAIN

ALWAYS ON DUTY

I need at least 2 cubes in BLOOD. Fewer? Say "foggy…" — and everybody wobbles like wet spaghetti for five seconds.

PANCREAS ALARM

EVERYONE ELSE

Shout ALARM! when BLOOD holds more than 5 cubes. Shout ALL CLEAR when it's 5 or fewer.

LEG MUSCLES

ROUND 3 ONLY

Grab cubes from BLOOD into LEG MUSCLES — you don't wait for the alarm.

LEG MUSCLES

ROUND 3 ONLY

Grab cubes from BLOOD into LEG MUSCLES — you don't wait for the alarm.

lumisnutritionlab.com · © 2026 Soyee Lee · Free to print and copy for home and classroom use.

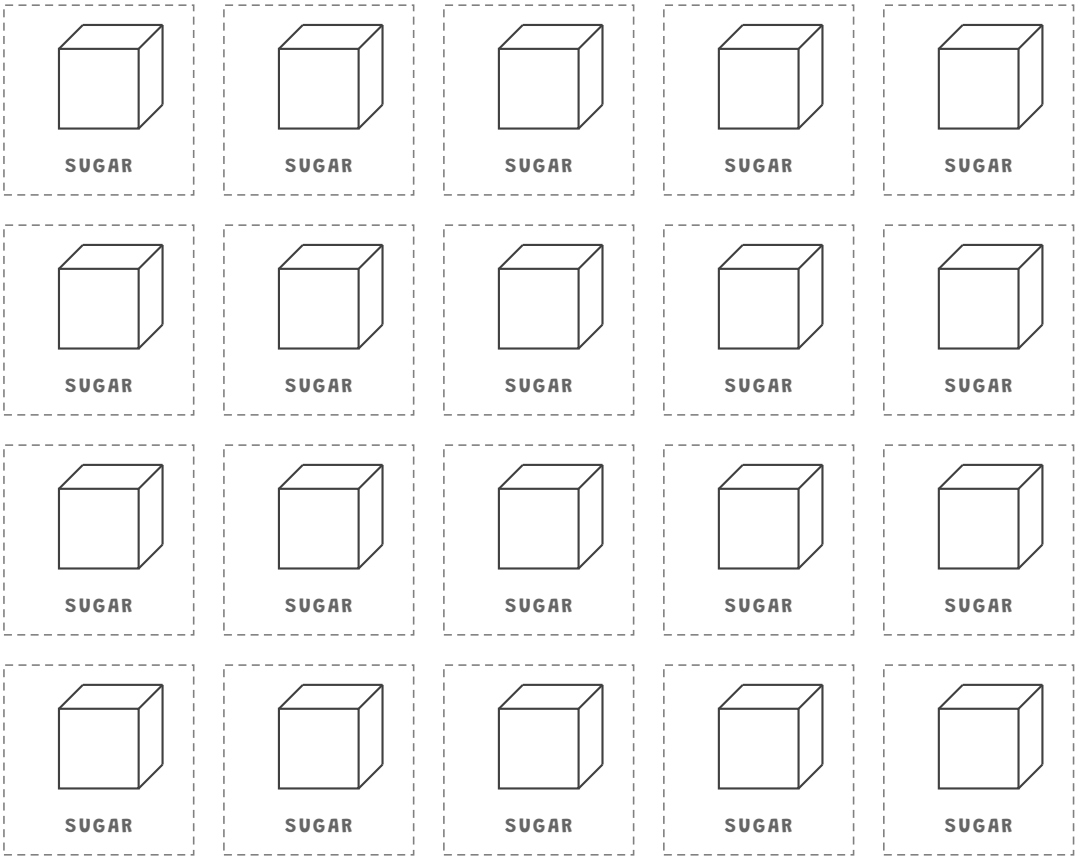
Signs for the boxes

BLOOD

CELLS

LEG MUSCLES

20 SUGAR CUBES



lumisnutritionlab.com · © 2026 Soyee Lee · Free to print and copy for home and classroom use.